

HAT-P-54b

R-1052

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-54_251009 · created 2026-07-30T10:14:09+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2460957.9341 +/- 0.006 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.139 +/- 0.057
Transit depth (Rp/Rs)^2	1.94 +/- 1.58 [%]
Orbital Inclination (inc)	89.27 +/- 2.31
Airmass coefficient 1 (a1)	1.03 +/- 0.13
Airmass coefficient 2 (a2)	-0.011 +/- 0.078
Scatter in the residuals of the lightcurve fit is	13.0 %
Best Comparison Star	#1 - [255, 198]
Optimal Method	PSF photometry
Transit Duration (day)	0.0686 +/- 0.0075

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.139 ± 0.057 vs 0.1572 (deviation 0.32σ)
Duration fit vs published	0.0686 d vs 0.075 d (deviation 0.85σ)
Mid-time O-C	13.8 min
Depth SNR	2.4
Residual scatter	13.0 %

Light curve

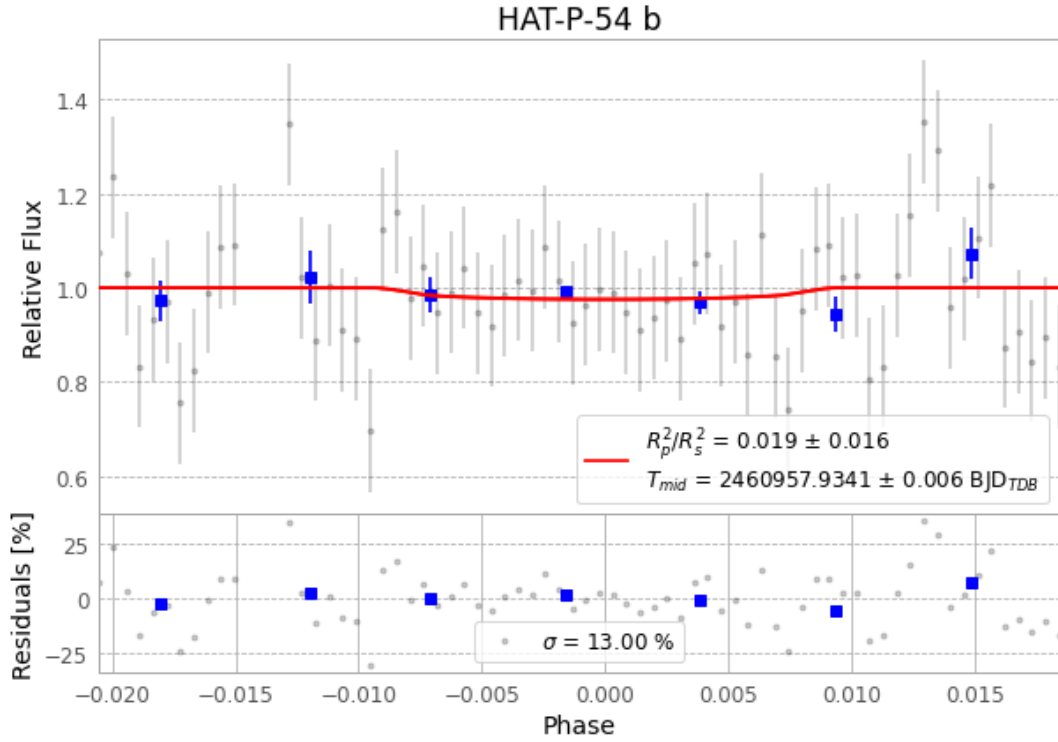


Figure 1 — FinalLightCurve_HAT-P-54 b_2025-10-09.png (SHA-256 724494e183ffa9d0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [514, 267], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 944ad5b1560a66d3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

72 FITS frames (47 MB), HATP-54251009083015.FITS → HATP-54251009120315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-54-251009.log (df86e395a342...), FinalLightCurve_HAT-P-54 b_2025-10-09.png (724494e183ff...), FinalLightCurve_HAT-P-54 b_2025-10-09.csv (7666614cf275...)

Compute resources

Wall time 135.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-30 10:14Z.